AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

Physics Paper II

Time: 2 hours 20 minutes Marks: 55

AKEB May 2018 Test for Teaching & Learning Only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are FOURTEEN questions. Answer ALL questions. Questions 12, 13 & 14 each offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Compare a battery to a capacitor in THREE points.

S. No.	Battery	Capacitor
1		
2		
3		

Q.2. (Total 4 Marks)

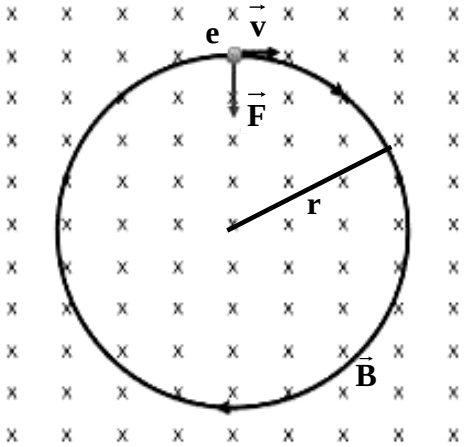
Two identical cells of electromotive force (e.m.f.) 2.0 V are joined together in parallel combination. They provide power to an external circuit consisting of two resistors of 12 Ω each which are joined in parallel combination with a circuit and a voltmeter which reads a voltage of 1.6 V.

Calculate the internal resistance of each cell and the power dissipated by the internal resistance.

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Q.3. (Total 5 Marks)

An electron of mass (m) and charge (e) is accelerated with a velocity ($\vec{v}$) by a potential difference (V) which is projected perpendicularly to a uniform magnetic field ($\vec{B}$).



The electron moves along a circular trajectory of radius (r).

a. Mention the expression for the

i. speed of the electron. (1 Mark)

_____

ii. ratio $\frac{e}{m}$. (1 Mark)

_____

b. Show that the radius of the circular trajectory of the electron is $r = \frac{1}{B}\sqrt{\frac{2Vm}{e}}$. (3 Marks)

_____

_____

_____

_____

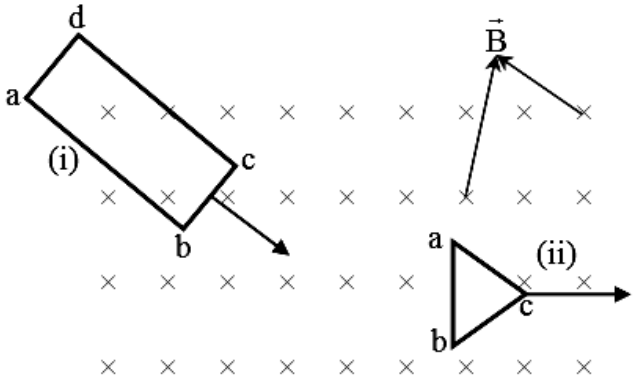
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Q.4. (Total 5 Marks)

The given figure shows a region of magnetic field ($\vec{B}$).



a. If a rectangular loop (abcd) is entering into the magnetic field, then the magnetic flux (Φ_B) through the rectangular loop (abcd) increases or decreases? Give reason to support your answer. (2 Marks)

b. A triangular loop (abc) is leaving the magnetic field. Is the induced current along (bacb) or (bcab)? Give reason to support your answer. (2 Marks)

c. If both the loops are completely static inside the region of the magnetic field, then there will be no induced current. True or False? (1 Mark)

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Q.5. (Total 2 Marks)

Define the following terms:

a. Peak value (voltage).

_____

_____

b. Root mean square (r.m.s.) value (voltage).

_____

_____

Q.6. (Total 3 Marks)

The given diagram shows three types of waves.

1 

2 

3 

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a. Identify amplitude modulation (AM) and frequency modulation (FM) waves. (2 Marks)

_____

_____

_____

_____

b. Define any one of the above identified waves. (1 Mark)

_____

_____

Q.7. (Total 3 Marks)

Give THREE differences between elastic and plastic deformations in solids.

S. No.	Elastic Deformation	Plastic Deformation
1		
2		
3		

Q.8. (Total 3 Marks)

a. Define a transistor. (1 Mark)

_____

_____

b. Mention TWO equations which are regarded as fundamental equations of all transistors. (2 Marks)

_____

_____

_____

_____

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Q.9. (Total 3 Marks)

a. Define the following:

i. Inertial frame of references (1 Mark)

ii. Non-inertial frame of references (1 Mark)

b. Give ONE example of a non-inertial frame of reference. (1 Mark)

Q.10. (Total 4 Marks)

Determine the wavelength of radiation emitted by a hydrogen atom when an electron jumps from n = 4 to the n = 2 orbit in the Bohr atomic model.

(**Note**: Given that Rydberg's constant = R = 1.0974×10^7 m^{-1}.)

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Q.11. (Total 5 Marks)

a. Describe the phenomenon of radioactive decay. (2 Marks)

b. Write ONE balanced equation for each of the following decays.

i. Alpha (α) (1 Mark)

ii. Beta (β) (1 Mark)

iii. Gamma (γ) (1 Mark)

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Q.12. (Total 5 Marks)

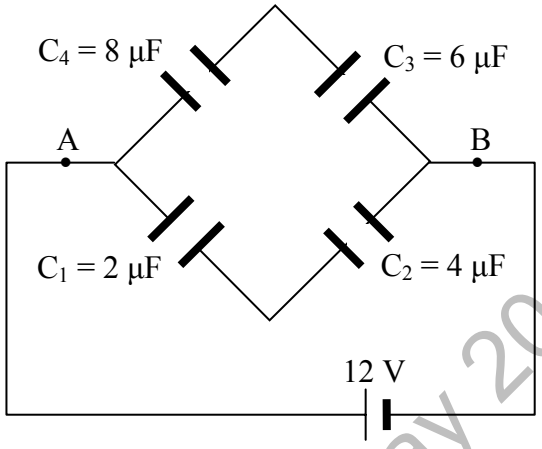
EITHER

a. Calculate the electric field intensity at midpoint between two charges + 1.6×10^{-19} C and -1.6×10^{-19} C, separated by a distance of 50 cm.

(**Note**: The value of the constant of proportionality 'k' = 9×10^9 Nm^2/C^2)

OR

b. The diagram represents a combination of different capacitors.



Calculate the effective capacitance between terminals (A) and (B).

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Q.13. (Total 5 Marks)

EITHER

a. Write any FIVE differences between inductance and induction.

S. No.	Inductance	Induction
1		
2		
3		
4		
5		

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OR

b. Write any FIVE differences between alternating and direct current generators.

S. No.	AC generator	DC generator
1		
2		
3		
4		
5		

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Q.14. (Total 5 Marks)

EITHER

a. Describe any FIVE conclusions that can be extracted by Einstein's special theory of relativity.

OR

b. Suppose a spacecraft is moving from the Earth to a distant star. A stationary observer is observing the spacecraft from the Earth. If the spacecraft and the observer have similar clocks, then according to the special theory of relativity, the clock on the spacecraft runs slower as compared to the observer's clock.

Elaborate the given scenario with the help of mathematical formulas.

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END OF PAPER

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